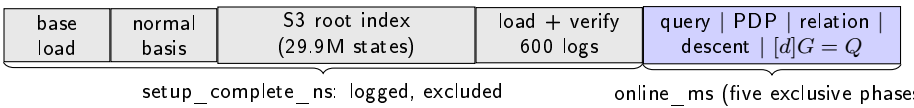
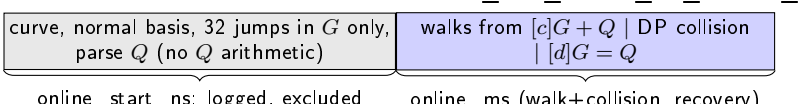
Once per job, precompute step (one pinned CPU, timed, excluded from online, charged in cold ratio and BL budget)

IC rank stage: 600 guided relations $[a]G - R_j$ (sequential), linear algebra, write factor-log table; independent $[L_j]G = R_j$ check in oracle.py

IC arm, every process (koblitz_orbit_dlp_fast_online + ic_log_cache.patch)



Strong rho arm, every process (koblitz_rho_batch_ks_strong_online, rung 3)



Legacy packed rho, Panel C only (koblitz_rho_fixture packed; not a vs_rho reference)

